Scalability & Future Plan

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Project Name: Health360: Comprehensive Healthcare Risk & Disease Analytics
Maximum Marks: 1 Mark

This document outlines how the current project solution can be scaled to handle larger user bases, increased data loads, or extended features in the future. It also captures the team's roadmap for enhancing and evolving the project beyond its current state.

Current System Limitations:

S.No	Limitation	Impact	Priority to Address (High / Medium / Low)
1	Single-node data processing and batch ETL workflows	Limited real-time stream ingestion throughput for large IoT wearable deployments	High
2	Static rule-based health risk scoring formula	Lower adaptability to evolving clinical disease markers and multi-variable correlations	Medium
3	Limited local/cloud instance caching	Potential latency spikes during peak concurrent clinician dashboard queries	Low

Scalability Plan:

S.No	Scalability Aspect	Current State	Proposed Upgrade / Solution
1	User Load	Supports ~500 concurrent users on standard web/mobile interface	Deploy Kubernetes container orchestration with auto-scaling load balancers to support 50,000+ concurrent active users.
2	Data Storage	Snowflake Staging & Curated tables hosting 5,000 patient records (~1.2 MB)	Implement distributed data lake architecture (AWS S3 + Snowflake Enterprise)

			supporting petabytes of multi-modal EHR & IoT streams.
3	Performance	Standard SQL queries and indexing (~0.12s - 0.38s execution time)	Integrate Redis caching layers, materialized views, and partitioned warehouse clustering for sub-millisecond query responses.
4	Security	Role-based access control and basic encryption in transit	Implement end-to-end Zero Trust security, HIPAA/GDPR compliance auditing, and hardware security modules (HSM).

Future Roadmap:

Phase	**Planned Feature / Enhancement**	**Target Timeline**	**Expected Impact**
Phase 2	Automated real-time IoT wearable sensor stream integration via MQTT and Kafka	Q1 2027	Enables live patient vital sign tracking and instant anomaly alerts.
Phase 3	Advanced Deep Learning models (LSTM / Transformer-based) for early sepsis and cardiac arrest prediction	Q2-Q3 2027	Increases disease risk forecasting accuracy and proactive clinical intervention.
Phase 4	Mobile app push notifications, telemedicine video consultations, and hospital EMR API interoperability (HL7/FHIR)	Q4 2027	Delivers a fully integrated, omni-channel ecosystem for patients and healthcare providers.